Aditya Singh

Data Analyst / Machine Learning Enthusiast

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Professional Summary

Information Technology undergraduate skilled in SQL, Python, Power BI, Excel, and Machine Learning. Experienced in building predictive models, data analysis dashboards, and recommendation engines. Proven leadership as a Google Student Ambassador and practical IT experience from an internship at Hindustan Aeronautics Limited (HAL). Seeking to apply analytical storytelling, statistical modeling, and machine learning to solve real-world business problems.

Technical Skills

- **Languages & Query Tools:** Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn, OpenCV), SQL (MySQL, SQL Server)
- **Data & Visualization:** Power BI, Microsoft Excel (Advanced formulas, Pivot Tables, Power Query, DAX), Tableau
- **Machine Learning & AI:** Regression, Classification, Clustering, Recommendation Systems, Model Evaluation, SMOTE
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code, SQL Server Management Studio (SSMS)

Experience

IT/Networking Intern – Hindustan Aeronautics Limited (HAL), Avionics Division 2 Jun 2026 – 30 Jun 2026

- Gained practical exposure to enterprise networking concepts including Ethernet, switches, routers, firewalls, and servers in defense IT.
- Studied LAN architecture, IP addressing (IPv4/IPv6), and subnetting; assisted in troubleshooting connectivity issues.
- Observed professional IT operations, service management workflows, and technical documentation standards in a defense environment.

Projects

Revenue Prediction | *Python, Scikit-learn, Pandas, NumPy* [GitHub] | [Colab]

- Built a regression-based ML model to forecast business revenue from historical sales data using feature engineering and hyperparameter tuning.
- Formulated and compared multiple algorithms (Linear Regression, Ridge, Random Forest Regressor) to select the best-performing model.
- Evaluated performance metrics (R^2, RMSE, MAE) and tuned hyperparameters via GridSearchCV to optimize prediction accuracy.

Netflix Recommender System | *Python, Scikit-learn, Pandas, NumPy* [GitHub] | [Colab]

- Developed a machine learning recommendation model for 8,000+ Netflix titles by analyzing movie metadata, genres, and ratings.
- Preprocessed unstructured data using Pandas and NumPy, performing missing value imputation, text cleaning, and feature extraction.
- Analyzed content distributions and release patterns to generate accurate, data-driven movie and TV show recommendations.

Credit Card Fraud Detection | *Python, Scikit-learn, Imbalanced-learn, SMOTE* [GitHub] | [Colab]

- Developed a classification model to detect fraudulent transactions in an imbalanced dataset of 280K+ records using SMOTE.
- Evaluated model performance using Confusion Matrix, Precision-Recall AUC, and F1-score to minimize false negatives.
- Achieved a 94% recall rate on fraud instances while optimizing decision thresholds for real-time transaction monitoring.

Education

Bachelor of Technology (B.Tech.) in Information Technology (IT) 2024 – 2028

Dr. Rammanohar Lohia Avadh University, Ayodhya, India

- **Relevant Coursework:** Database Management Systems (DBMS), Data Structures & Algorithms (DSA), Computer Networks, Operating Systems (OS), Object-Oriented Programming (OOPs), Python, SQL.

Certifications & Achievements

- **SQL Certification** – Intellipaat 2025
- **HAL Internship Certificate** – Hindustan Aeronautics Limited 1 July 2026
- **SIH 2026 Internal Hackathon – Selected** (Team ZERO2PROD) – PRAGYAN, IET RMLAU 19 Sep 2026